Learning Objective 7.4: I can apply the magnetic field boundary conditions to calculate magnetic fields on both sides of a material boundary.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate). In every problem, $\hat{\mathbf{n}}$ points from region 2 into region 1.

1. **Deriving the conditions.** Two magnetic materials meet at a flat boundary that carries a surface current density $\mathbf{J}_s$.

   (a) Apply Ampere's law to a rectangular loop with long sides of length W along a tangent $\hat{\mathbf{t}}$, one in each region, and short sides of height $h \to 0$. Show that $H_{t1} - H_{t2} = \mathbf{J}_s \cdot \left(\hat{\mathbf{n}} \times \hat{\mathbf{t}}\right)$.

   (b) Apply Gauss's law for magnetism to a pillbox with faces of area ΔA and height $h \to 0$. Show that $B_{n1} = B_{n2}$, and explain why no source term appears.

2. **Into a magnetic material.** The plane $z = 0$ separates air, region 1 at $z > 0$, from a material with $\mu_r = 500$, region 2 at $z < 0$. No surface current flows on the boundary. Just above it, $\mathbf{H}_1 = \hat{\mathbf{x}}10 + \hat{\mathbf{z}}20$ A/m.

   (a) Find $\mathbf{H}_2$ just below the boundary.

   $\mathbf{H}_2 =$ [ ]

   (b) Find $\mathbf{B}_1$ and $\mathbf{B}_2$.

   $B_z =$ [ ]

   (c) Find the angle each field makes with the normal, and check the ratio of their tangents against μ_1/μ_2.

   $\theta_1 =$ [ ]

   $\theta_2 =$ [ ]

   (d) In which region is $|\mathbf{H}|$ larger, and in which is $|\mathbf{B}|$ larger? Explain in one sentence each.

3. **A boundary that carries current.** The plane $z = 0$ separates region 1, $z > 0$, with $\mu = \mu_0$, from region 2, $z < 0$, with $\mu = 3\mu_0$. The boundary carries $\mathbf{J}_s = \hat{\mathbf{y}}4$ A/m. Just below it, $\mathbf{H}_2 = \hat{\mathbf{x}}2 - \hat{\mathbf{y}} + \hat{\mathbf{z}}6$ A/m.

   (a) Find $\mathbf{H}_1$.

   $\mathbf{H}_1$ =

   (b) Check your answer by computing $\hat{\mathbf{n}} \times (\mathbf{H}_1 - \mathbf{H}_2)$.

4. **A current sheet and an iron core.**

   (a) A sheet of current $\mathbf{J}_s = \hat{\mathbf{y}}20$ A/m lies in the plane $x = 0$ in free space. Lesson 30 found that it makes a field of magnitude $J_s/2$ on each side, parallel to the sheet and at right angles to the current. Find $\mathbf{H}$ on each side, and show that the tangential condition holds.

   (b) Inside a ferrite core with $\mu_r = 2000$, the field meets the core's surface $85°$ from the normal. Find the angle from the normal in the air just outside, and explain why a high-permeability core guides magnetic field.

Documentation: _____